Sources: Sheldon Axler, *Measure, Integration & Real Analysis* (MIRA) and its *Supplement*, at measure.axler.net.

Exercise 1.1 (MIRA 2A.4). Suppose F is a subset of $\mathbf{R}$ with the property that every open cover of F has a finite subcover. Prove that F is closed and bounded.

Solution 1.1. If $F = \emptyset$, then F is closed and bounded. Thus assume $F \neq \emptyset$. First we show that F is bounded. By the Archimedean Property, every $x \in \mathbf{R}$ satisfies $|x| < n$ for some $n \in \mathbf{Z}^+$. Thus the open intervals $(-n, n)$, for $n \in \mathbf{Z}^+$, are open sets that cover F, and by hypothesis finitely many of them cover F. Let N be the largest index among them. Because the intervals increase with n, we have $F \subseteq (-N, N)$. Hence $|y| < N$ for all $y \in F$, and F is bounded.

To show that F is closed, suppose $x \in \mathbf{R} \setminus F$. For $n \in \mathbf{Z}^+$, let

$$G_n = \left(-\infty,\, x - \tfrac{1}{n}\right) \cup \left(x + \tfrac{1}{n},\, \infty\right),$$

which is open because each of the two intervals is open and a union of open sets is open. If $y \in F$, then $y \neq x$, and thus $|y - x| > \frac{1}{n}$ for some $n \in \mathbf{Z}^+$ by the Archimedean Property, which means that $y \in G_n$. Thus $\{G_n : n \in \mathbf{Z}^+\}$ is an open cover of F, and finitely many G_n cover F. Let N be the largest index among them. Because $G_1 \subseteq G_2 \subseteq \cdots$, we have $F \subseteq G_N$, which implies that

$$\left(x - \tfrac{1}{N},\, x + \tfrac{1}{N}\right) \subseteq \mathbf{R} \setminus F.$$

Thus $\mathbf{R} \setminus F$ is open, and hence F is closed, completing the proof. □

Exercise 1.2 (MIRA 2A.6). Prove that if $a, b \in \mathbf{R}$ and $a < b$, then $|(a, b)| = |[a, b)| = |(a, b]| = b - a$.

Solution 1.2. Because outer measure preserves order (2.5) and the outer measure of a closed interval is its length (2.14), the inclusions $(a, b) \subseteq [a, b) \subseteq [a, b]$ and $(a, b) \subseteq (a, b] \subseteq [a, b]$ give

$$|(a, b)| \leq |[a, b)| \leq b - a \qquad \text{and} \qquad |(a, b)| \leq |(a, b]| \leq b - a.$$

Because finite sets have outer measure 0 (Example 2.3), countable subadditivity of outer measure (2.8) applied to $[a, b] = (a, b) \cup \{a, b\}$ gives

$$b - a = |[a, b]| \leq |(a, b)| + |\{a, b\}| = |(a, b)|.$$

Thus $b - a \leq |(a, b)| \leq |[a, b)| \leq b - a$, and the same holds with $(a, b]$ in place of $[a, b)$. Hence all three outer measures equal $b - a$. □

Exercise 1.3 (MIRA 2B.3). Suppose $\mathcal{S}$ is the smallest σ-algebra on $\mathbf{R}$ containing $\{(r, s] : r, s \in \mathbf{Q}\}$. Prove that $\mathcal{S}$ is the collection of Borel subsets of $\mathbf{R}$.

Solution 1.3. Let $\mathcal{A} = \{(r, s] : r, s \in \mathbf{Q}\}$, and let $\mathcal{B}$ be the collection of Borel subsets of $\mathbf{R}$, which is the smallest σ-algebra on $\mathbf{R}$ containing every open set (2.29). Thus $\mathcal{S} \subseteq \mathcal{B}$ if $\mathcal{A} \subseteq \mathcal{B}$, and $\mathcal{B} \subseteq \mathcal{S}$ if every open set is in $\mathcal{S}$, by the definition of the smallest σ-algebra containing a collection (2.27).

First suppose $r, s \in \mathbf{Q}$. A real number t satisfies $t \leq s$ if and only if $t < s + \frac{1}{k}$ for every $k \in \mathbf{Z}^+$, because if $t > s$, then the Archimedean Property gives $k \in \mathbf{Z}^+$ with $s + \frac{1}{k} < t$. Hence

$$(r, s] = \bigcap_{k=1}^{\infty} \left(r,\, s + \tfrac{1}{k}\right).$$

Each $(r, s + \frac{1}{k})$ is an open set and thus in $\mathcal{B}$, and $\mathcal{B}$ is closed under countable intersections (2.25(c)). Thus $(r, s] \in \mathcal{B}$, and $\mathcal{A} \subseteq \mathcal{B}$.

Now suppose $G \subseteq \mathbf{R}$ is open and $x \in G$. There exists $\delta > 0$ such that $(x - \delta, x + \delta) \subseteq G$, and there exist $r, s \in \mathbf{Q}$ with $x - \delta < r < x < s < x + \delta$, because there is a rational number between any two real numbers. Then $x \in (r, s] \subseteq G$. Because $\mathbf{Q}$ is countable, there is a sequence $q_1, q_2, \ldots$ containing every rational number. Listing the pairs (q_i, q_j) with $i + j = 2$, then those with $i + j = 3$, and so on, gives a sequence $(r_1, s_1), (r_2, s_2), \ldots$ containing every element of $\mathbf{Q} \times \mathbf{Q}$. For $k \in \mathbf{Z}^+$, let $A_k = (r_k, s_k]$ if $(r_k, s_k] \subseteq G$, and $A_k = \emptyset$ otherwise. Because $\emptyset = (0, 0] \in \mathcal{A}$, each A_k is in $\mathcal{A} \subseteq \mathcal{S}$. Each A_k is contained in G, and the interval $(r, s]$ above, which contains x, is one of the A_k. Thus $G = \bigcup_{k=1}^{\infty} A_k$, and hence $G \in \mathcal{S}$ because $\mathcal{S}$ is a σ-algebra. Hence $\mathcal{B} \subseteq \mathcal{S}$, and thus $\mathcal{S} = \mathcal{B}$, as desired. □

Exercise 1.4 (MIRA 2B.13). Suppose $(X, \mathcal{S})$ is a measurable space, $E_1, \ldots, E_n$ are disjoint subsets of X, and $c_1, \ldots, c_n$ are distinct nonzero real numbers. Prove that $c_1\chi_{E_1} + \cdots + c_n\chi_{E_n}$ is an $\mathcal{S}$-measurable function if and only if $E_1, \ldots, E_n \in \mathcal{S}$.

Solution 1.4. Let $f = c_1\chi_{E_1} + \cdots + c_n\chi_{E_n}$, a function from X to $\mathbf{R}$.

First suppose $E_1, \ldots, E_n \in \mathcal{S}$. Each χ_{E_k} is $\mathcal{S}$-measurable because $E_k \in \mathcal{S}$ (Example 2.38). Each constant function is $\mathcal{S}$-measurable because the inverse image of every set under it is $\emptyset$ or X, both of which are in $\mathcal{S}$. Thus each $c_k\chi_{E_k}$, and then their sum f, is $\mathcal{S}$-measurable, because sums and products of $\mathcal{S}$-measurable functions are $\mathcal{S}$-measurable (2.46(a)).

Now suppose f is $\mathcal{S}$-measurable, and suppose $k \in \{1, \ldots, n\}$. Because $E_1, \ldots, E_n$ are disjoint, f equals c_j on E_j for each j and equals 0 on $X \setminus (E_1 \cup \cdots \cup E_n)$. Because $c_1, \ldots, c_n$ are distinct and nonzero, this implies that $f(x) = c_k$ for all $x \in E_k$ and $f(x) \neq c_k$ for all $x \notin E_k$. Thus $f^{-1}(\{c_k\}) = E_k$. The set $\{c_k\}$ is closed and hence a Borel set (Example 2.30). Thus $E_k = f^{-1}(\{c_k\}) \in \mathcal{S}$ by the definition of an $\mathcal{S}$-measurable function (2.35). Hence $E_1, \ldots, E_n \in \mathcal{S}$. □

Exercise 1.5 (MIRA 2B.27). Prove or give a counterexample: If $(X, \mathcal{S})$ is a measurable space and $f : X \to [-\infty, \infty]$ is a function such that $f^{-1}\big((a, \infty)\big) \in \mathcal{S}$ for every $a \in \mathbf{R}$, then f is an $\mathcal{S}$-measurable function.

Solution 1.5. The statement is false. Let $X = \{p, q\}$ with $p \neq q$, and let $\mathcal{S} = \{\emptyset, X\}$, which is a σ-algebra on X. Define $f : X \to [-\infty, \infty]$ by $f(p) = \infty$ and $f(q) = -\infty$.

If $a \in \mathbf{R}$, then (a, ∞) contains neither ∞ nor $-\infty$, and thus $f^{-1}\big((a, \infty)\big) = \emptyset \in \mathcal{S}$. Hence f satisfies the hypothesis.

However, $\{\infty\} \cap \mathbf{R} = \emptyset$ is a Borel set, and thus $\{\infty\}$ is a Borel subset of $[-\infty, \infty]$ by the definition of Borel subsets of $[-\infty, \infty]$ (2.50). Because $f^{-1}(\{\infty\}) = \{p\} \notin \mathcal{S}$, the function f is not $\mathcal{S}$-measurable, by the definition of a measurable function with values in $[-\infty, \infty]$ (2.51). □

Exercise 1.6 (MIRA 2C.1). Explain why there does not exist a measure space $(X, \mathcal{S}, \mu)$ with the property that $\{\mu(E) : E \in \mathcal{S}\} = [0, 1)$.

Solution 1.6. Suppose $(X, \mathcal{S}, \mu)$ is a measure space. Then $X \in \mathcal{S}$ because every σ-algebra on X contains X (2.25(a)), and every $E \in \mathcal{S}$ satisfies $E \subseteq X$ and hence $\mu(E) \leq \mu(X)$, because measures preserve order (2.57(a)). Thus $\mu(X)$ is the largest element of $\{\mu(E) : E \in \mathcal{S}\}$. But

$[0,1)$ has no largest element: if $t \in [0,1)$, then $\frac{1}{2}(t+1) \in [0,1)$ and $\frac{1}{2}(t+1) > t$. Hence $\{\mu(E) : E \in \mathcal{S}\} \neq [0,1)$. □

Exercise 1.7 (MIRA 2C.5). Suppose $(X, \mathcal{S}, \mu)$ is a measure space such that $\mu(X) < \infty$. Prove that if $\mathcal{A}$ is a set of disjoint sets in $\mathcal{S}$ such that $\mu(A) > 0$ for every $A \in \mathcal{A}$, then $\mathcal{A}$ is a countable set.

Solution 1.7. For $n \in \mathbf{Z}^+$, let $\mathcal{A}_n = \{A \in \mathcal{A} : \mu(A) > \frac{1}{n}\}$. Suppose $m \in \mathbf{Z}^+$ and $A_1, \ldots, A_m$ are distinct elements of $\mathcal{A}_n$. They are disjoint sets in $\mathcal{S}$. Thus finite additivity of μ, which follows from the definition of measure (2.54), and order preservation (2.57(a)) give

$$\frac{m}{n} < \mu(A_1) + \cdots + \mu(A_m) = \mu(A_1 \cup \cdots \cup A_m) \leq \mu(X).$$

Hence $m < n\mu(X) < \infty$. Thus $\mathcal{A}_n$ has at most $n\mu(X)$ elements and is finite.

Now suppose $A \in \mathcal{A}$. Then $0 < \mu(A) \leq \mu(X) < \infty$ because measures preserve order (2.57(a)), and the Archimedean Property gives $n \in \mathbf{Z}^+$ with $\frac{1}{n} < \mu(A)$. Thus $A \in \mathcal{A}_n$. Hence $\mathcal{A} = \bigcup_{n=1}^{\infty} \mathcal{A}_n$.

If $\mathcal{A} = \emptyset$, then $\mathcal{A}$ is countable. Thus assume $\mathcal{A} \neq \emptyset$. Because $\mathcal{A}_1 \subseteq \mathcal{A}_2 \subseteq \cdots$ and each $\mathcal{A}_n$ is finite, listing the elements of $\mathcal{A}_1$, then those of $\mathcal{A}_2 \setminus \mathcal{A}_1$, and so on, produces a sequence containing every element of $\mathcal{A}$. Thus $\mathcal{A}$ is countable. □

Exercise 1.8 (MIRA 2D.5). Prove that if $A \subseteq \mathbf{R}$ is Lebesgue measurable, then there exists an increasing sequence $F_1 \subseteq F_2 \subseteq \cdots$ of closed sets contained in A such that

$$\left| A \setminus \bigcup_{k=1}^{\infty} F_k \right| = 0.$$

Solution 1.8. Because A is Lebesgue measurable, the equivalences for being a Lebesgue measurable set (2.71) give closed sets $F_1, F_2, \ldots$ contained in A such that $\left|A \setminus \bigcup_{k=1}^{\infty} F_k\right| = 0$. For $k \in \mathbf{Z}^+$, let $F'_k = F_1 \cup \cdots \cup F_k$. Then $F'_1 \subseteq F'_2 \subseteq \cdots$, and each F'_k is a subset of A that is closed because a finite union of closed sets is closed. Because $F_k \subseteq F'_k \subseteq \bigcup_{j=1}^{\infty} F_j$ for every k, we have $\bigcup_{k=1}^{\infty} F'_k = \bigcup_{k=1}^{\infty} F_k$. Hence

$$\left| A \setminus \bigcup_{k=1}^{\infty} F'_k \right| = \left| A \setminus \bigcup_{k=1}^{\infty} F_k \right| = 0,$$

and $F'_1 \subseteq F'_2 \subseteq \cdots$ is the desired sequence. □

Exercise 1.9 (MIRA 2D.7). Prove that if $A \subseteq \mathbf{R}$ is Lebesgue measurable, then there exists a decreasing sequence $G_1 \supseteq G_2 \supseteq \cdots$ of open sets containing A such that

$$\left| \left(\bigcap_{k=1}^{\infty} G_k \right) \setminus A \right| = 0.$$

Solution 1.9. Suppose A is Lebesgue measurable. By the equivalences for being a Lebesgue measurable set (2.71), there exist open sets $G_1, G_2, \ldots$ containing A such that $\left|\left(\bigcap_{k=1}^{\infty} G_k\right) \setminus A\right| = 0$. For $k \in \mathbf{Z}^+$, let $G'_k = G_1 \cap \cdots \cap G_k$, which contains A and is open because a finite intersection of open sets is open. Then $G'_1 \supseteq G'_2 \supseteq \cdots$, and $\bigcap_{j=1}^{\infty} G_j \subseteq G'_k \subseteq G_k$ for every k, which implies

that $\bigcap_{k=1}^{\infty} G'_k = \bigcap_{k=1}^{\infty} G_k$. Thus $G'_1 \supseteq G'_2 \supseteq \cdots$ is a decreasing sequence of open sets containing A with

$$\left|\left(\bigcap_{k=1}^{\infty} G'_k\right) \setminus A\right| = \left|\left(\bigcap_{k=1}^{\infty} G_k\right) \setminus A\right| = 0. \qquad \square$$

Exercise 1.10 (MIRA 2D.10). Prove that if A and B are disjoint subsets of $\mathbf{R}$ and B is Lebesgue measurable, then $|A \cup B| = |A| + |B|$.

Solution 1.10. Countable subadditivity of outer measure (2.8) gives $|A \cup B| \leq |A| + |B|$. To prove the inequality in the other direction, recall that B is Lebesgue measurable. Thus there exists a Borel set $B' \subseteq B$ such that $|B \setminus B'| = 0$, by the definition of Lebesgue measurable set (2.70). Then $|B'| \leq |B|$ because outer measure preserves order (2.5), and countable subadditivity (2.8) applied to $B = B' \cup (B \setminus B')$ gives $|B| \leq |B'| + |B \setminus B'| = |B'|$. Thus $|B'| = |B|$.

The sets A and B' are disjoint because $B' \subseteq B$ and $A \cap B = \emptyset$, and B' is a Borel set. Thus additivity of outer measure if one of the sets is a Borel set (2.66) gives $|A \cup B'| = |A| + |B'|$. Because $A \cup B' \subseteq A \cup B$, order preservation (2.5) now gives

$$|A \cup B| \geq |A \cup B'| = |A| + |B'| = |A| + |B|. \qquad \square$$